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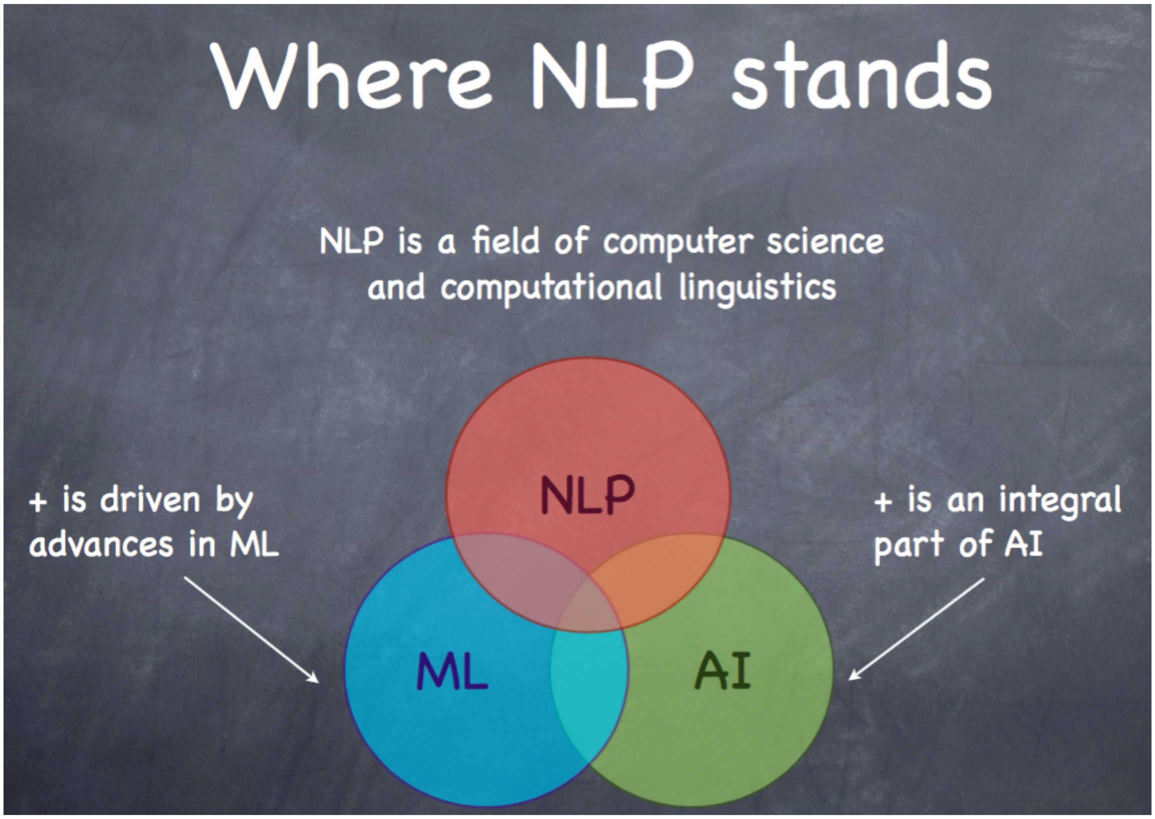
Introduction to Natural Language processing

Natural language processing (NLP): Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.



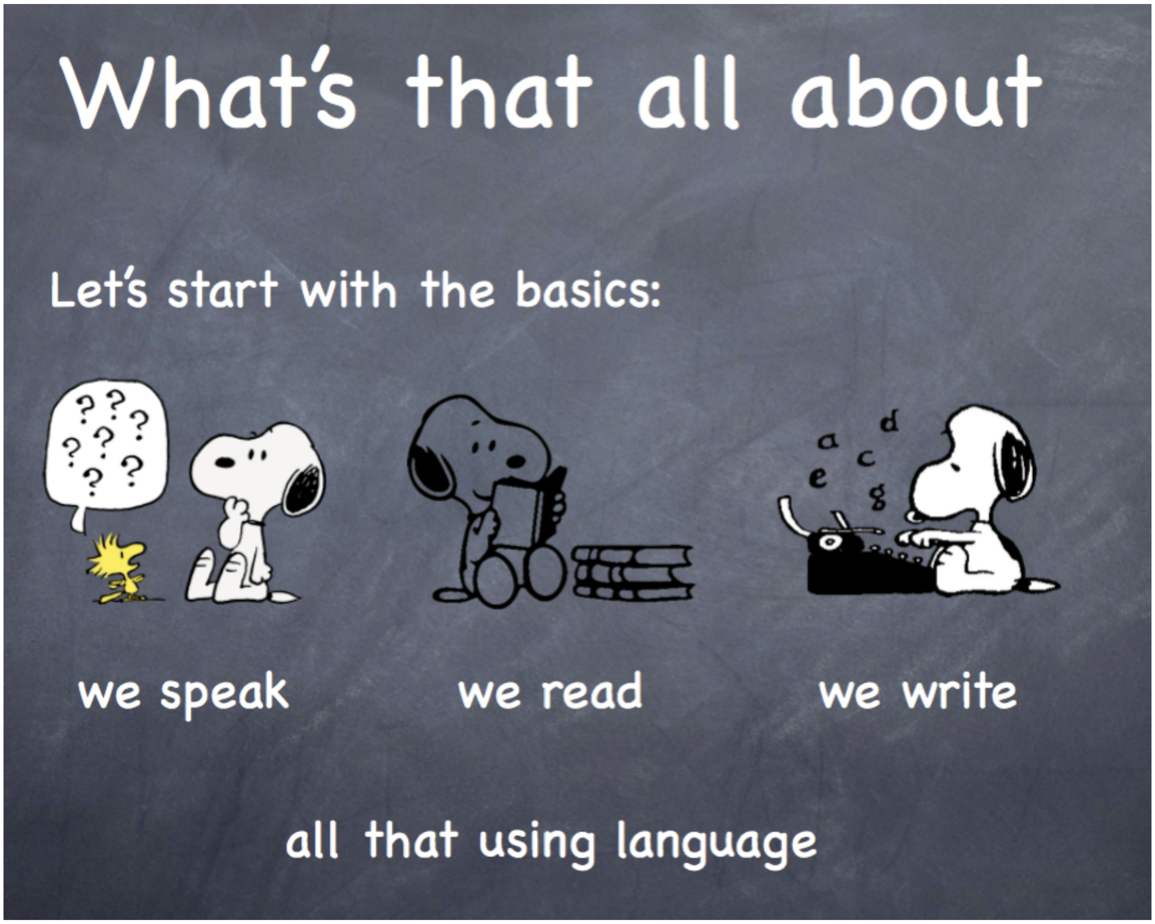
Where NLP stands

NLP is a field of computer science and computational linguistics

+ is driven by advances in ML

+ is an integral part of AI

What is it all about?

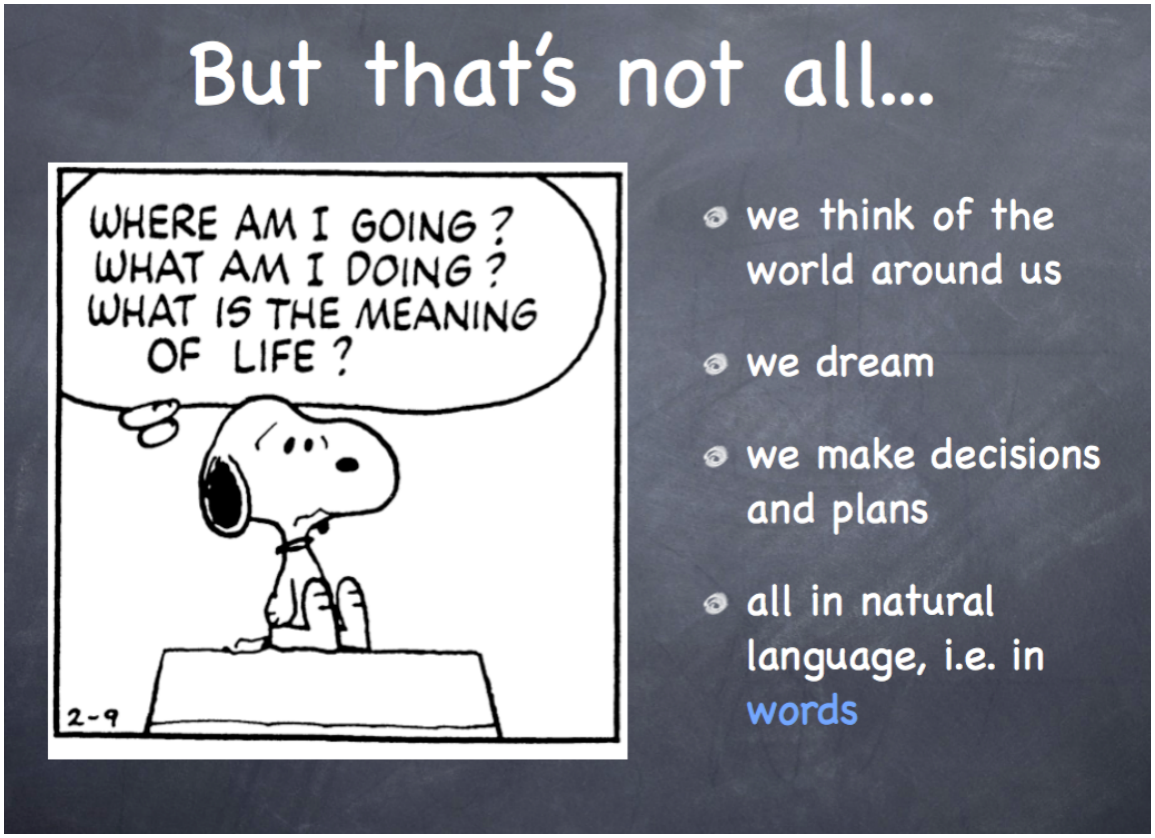


What's that all about

Let's start with the basics:

we speak we read we write

all that using language



But that's not all...

WHERE AM I GOING ? WHAT AM I DOING ? WHAT IS THE MEANING OF LIFE ?
2-9

- we think of the world around us
- we dream
- we make decisions and plans
- all in natural language, i.e. in words

What are the raw materials for NLP revolution?

1. Data: There is a huge amount of human knowledge that has been digitized.

2. Compute: Our personal computers are very powerful, and it has never been easier to rent more compute power.

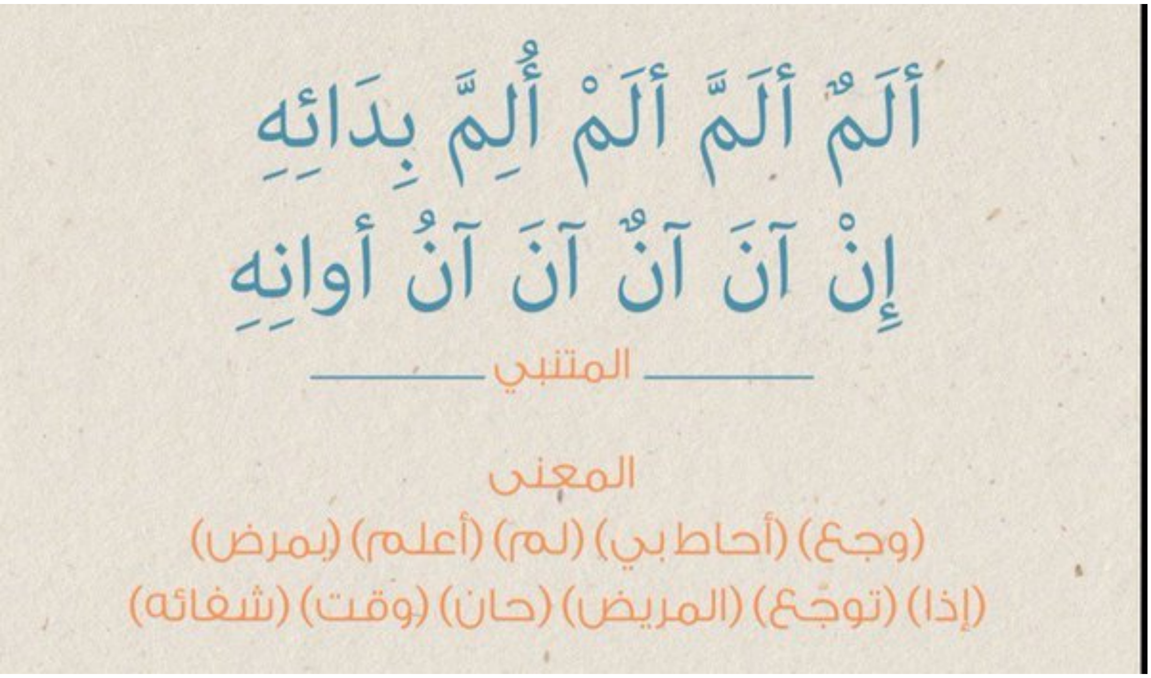
3. Algorithms: We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

Why NLP is hard?

It’s the nature of the human language that makes NLP difficult.

Some examples:

1. Ambiguity: One word/sentence can have different meanings.



أَلَمٌ أَلَمَّ أَلَمْ أُلِمَّ بِدَائِهِ
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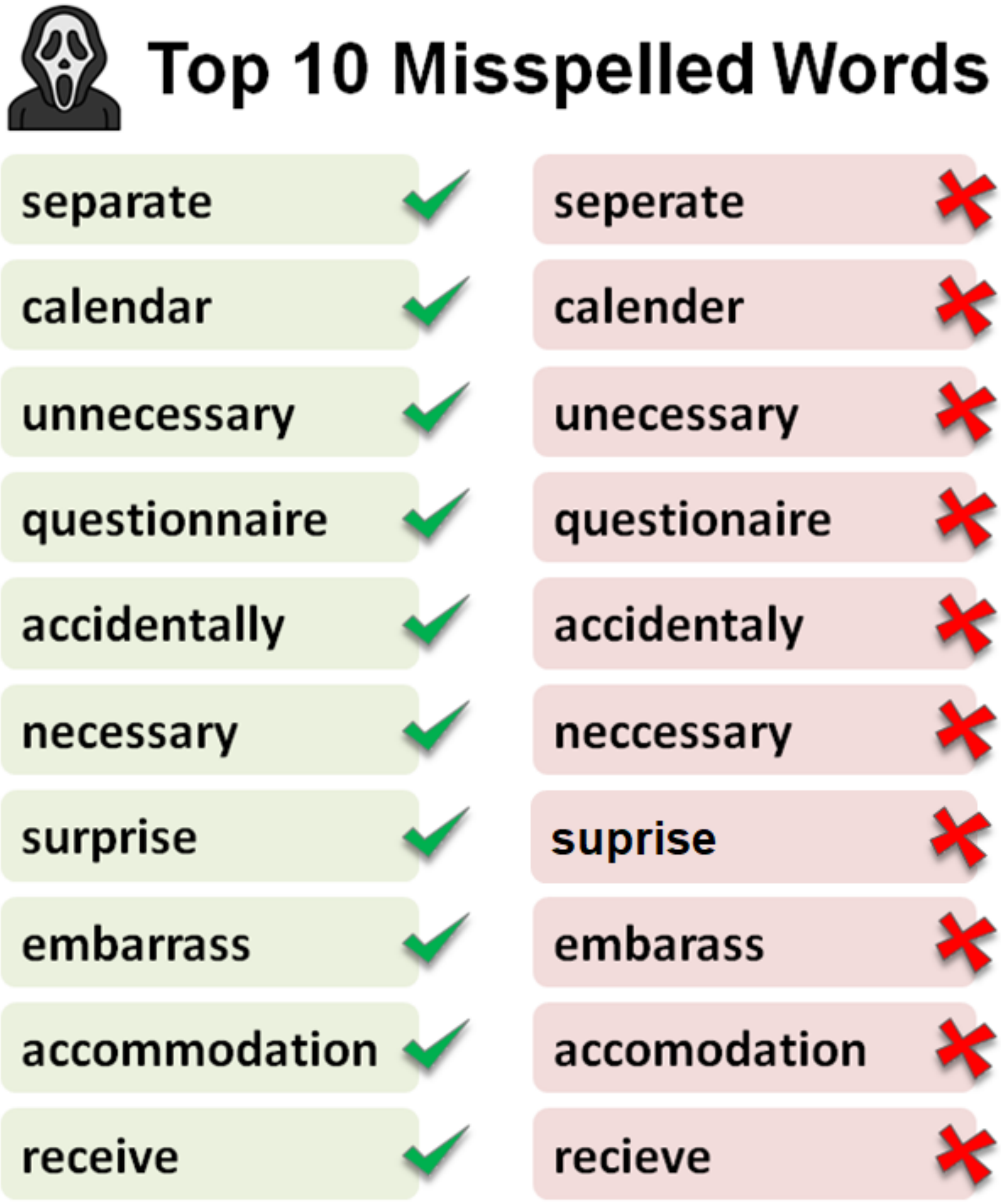
المتنبي

المعنى

(وجع) (أحاط بي) (لم) (أعلم) (بمرض)
(إذا) (توجّع) (المريض) (حان) (وقت) (شفائه)

2. Spelling Errors: There are some words that are usually misspelled by people.



Top 10 Misspelled Words

separate ✓	seperate ✗
calendar ✓	calender ✗
unnecessary ✓	unecessary ✗
questionnaire ✓	questionaire ✗
accidentally ✓	accidentaly ✗
necessary ✓	neccessary ✗
surprise ✓	suprise ✗
embarrass ✓	embarass ✗
accommodation ✓	accomodation ✗
receive ✓	recieve ✗

3. Phonetics and Phonology: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

4. Order of Context: "Dog bites man" and "Man bites dog" have vastly different meanings

In ML terms, why is NLP hard?

1. Input: Large, discrete, open-ended input state spaces

2. Output: Variable length output spaces

3. Algorithms: Recycled from other fields

Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF
- DOCX

NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

NLTK

Natural Language Toolkit (nltk) is a Python package for NLP https://www.nltk.org/

What are Corpus / Corpora?

a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
In [1]: !pip install nltk
```

```
Collecting nltk
  Downloading nltk-3.10.3-py3-none-any.whl.metadata (3.2 kB)
Collecting defusedxml (from nltk)
  Downloading defusedxml-0.7.1-py2.py3-none-any.whl.metadata (32 kB)
Collecting click (from nltk)
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting joblib (from nltk)
  Downloading joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting regex>=2021.8.3 (from nltk)
  Downloading regex-2026.9.10-cp313-cp313-win_amd64.whl.metadata (41 kB)
Collecting tqdm (from nltk)
  Downloading tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Collecting cloudpickle>=3.0 (from joblib->nltk)
  Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
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on313\site-packages (from tqdm->nltk) (0.4.6)
Downloading nltk-3.10.3-py3-none-any.whl (1.8 MB)
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Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Downloading joblib-1.6.0-py3-none-any.whl (306 kB)
Downloading cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Downloading tqdm-4.70.1-py3-none-any.whl (80 kB)
Installing collected packages: tqdm, regex, defusedxml, cloudpickle, click, joblib,
nltk

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Successfully installed click-8.5.0 cloudpickle-3.1.2 defusedxml-0.7.1 joblib-1.6.0 n
ltk-3.10.3 regex-2026.9.10 tqdm-4.70.1

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [4]:
```
import nltk
```

Arabic Libraries (Extra)

Camel-Tools: https://camel-tools.readthedocs.io/en/latest/

PyArabic: https://pyarabic.readthedocs.io/ar/latest/

Farasa: https://farasa.qcri.org/

NLP Applications

1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

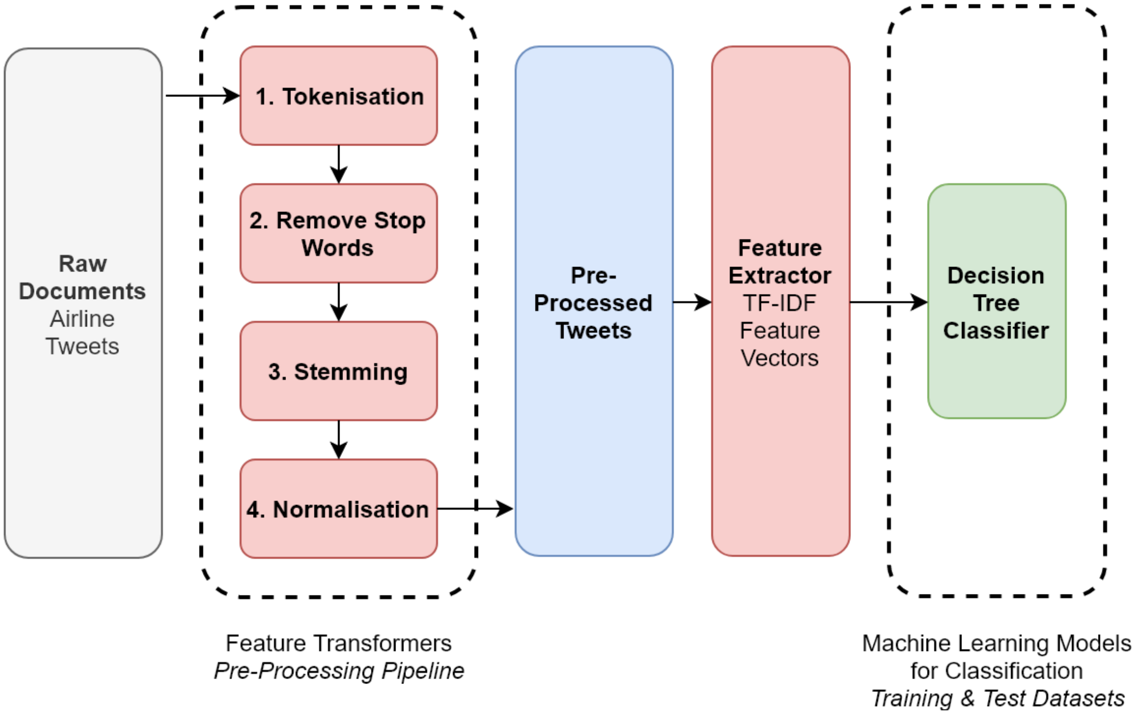
2. Machine Translation

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

3. Sentiment Analysis

Identifying sentiments and opinions stated in a text.

Example of an NLP Pipeline


Raw Documents
Airline Tweets
1. Tokenisation
2. Remove Stop Words
3. Stemming
4. Normalisation
Pre-Processed Tweets
Feature Extractor
TF-IDF Feature Vectors
Decision Tree Classifier
Feature Transformers
Pre-Processing Pipeline
Machine Learning Models for Classification
Training & Test Datasets

1. Text Acquisition:

Obtain the text data from a source, such as a file, a web page, or a database.

2. Text Preprocessing:

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all tokens to lowercase to ensure uniformity.
- **Removing Punctuation:** Strip away punctuation marks from tokens.
- **Stopword Removal:** Eliminate common words (e.g., "and," "the," "is") that may not carry significant meaning.
- **Stemming:** Removes prefixes/suffixes to get a root form. (Running -> run)
- **Lemmatization:** Converts word to base/dictionary form (lemma). (better -> good).

3. Feature Extraction:

- **Bag-of-Words (BoW):** Create a vector representation of the text using the frequency of each token in the document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Assign weights to tokens based on their importance in the document relative to the entire corpus.

4. Model Building:

Choose a machine learning model for sentiment analysis, such as a Naive Bayes classifier, Support Vector Machine (SVM), or a neural network.

Task#1:

Students should work in groups and search online for one of the different types of datasets.
• CSV • Text Files • JSON • SQL • HTML • PDF • DOCX

Task#2:

Download and open the dataset.

In [3]:
```
%pip install kagglehub
```

```
Collecting kagglehub
  Downloading kagglehub-1.0.2-py3-none-any.whl.metadata (40 kB)
Collecting kagglesdk<1.0,>=0.1.22 (from kagglehub)
  Downloading kagglesdk-0.1.37-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: packaging in C:\Users\user\AppData\Roaming\Python\Pyt
hon313\site-packages (from kagglehub) (26.0)
Collecting pyyaml (from kagglehub)
  Downloading pyyaml-6.0.3-cp313-cp313-win_amd64.whl.metadata (2.4 kB)
Collecting requests (from kagglehub)
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: tqdm in c:\Users\user\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from kagglehub) (4.70.1)
Collecting protobuf (from kagglesdk<1.0,>=0.1.22->kagglehub)
  Downloading protobuf-7.36.1-cp310-abi3-win_amd64.whl.metadata (595 bytes)
Collecting charset_normalizer<4,>=2 (from requests->kagglehub)
  Downloading charset_normalizer-3.5.1-cp313-cp313-win_amd64.whl.metadata (46 kB)
Collecting idna<4,>=2.5 (from requests->kagglehub)
  Downloading idna-3.19-py3-none-any.whl.metadata (9.2 kB)
Collecting urllib3<3,>=1.26 (from requests->kagglehub)
  Downloading urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Collecting certifi>=2023.5.7 (from requests->kagglehub)
  Downloading certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: colorama in C:\Users\user\AppData\Roaming\Python\Pyth
on313\site-packages (from tqdm->kagglehub) (0.4.6)
Downloading kagglehub-1.0.2-py3-none-any.whl (70 kB)
Downloading kagglesdk-0.1.37-py3-none-any.whl (262 kB)
Downloading protobuf-7.36.1-cp310-abi3-win_amd64.whl (456 kB)
Downloading pyyaml-6.0.3-cp313-cp313-win_amd64.whl (154 kB)
Downloading requests-2.34.2-py3-none-any.whl (73 kB)
Downloading charset_normalizer-3.5.1-cp313-cp313-win_amd64.whl (199 kB)
Downloading idna-3.19-py3-none-any.whl (68 kB)
Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
Installing collected packages: urllib3, pyyaml, protobuf, idna, charset_normalizer,
certifi, requests, kagglesdk, kagglehub

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b-1.0.2 kagglesdk-0.1.37 protobuf-7.36.1 pyyaml-6.0.3 requests-2.34.2 urllib3-2.7.0
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [5]:

```python
import kagglehub

# IMDB Dataset of 50K Movie Reviews
path = kagglehub.dataset_download("lakshmi25npathi/imdb-dataset-of-50k-movie-review

print("Path to dataset files:", path)
```

```
c:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\tqdm\auto.p
y:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See ht
tps://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
Downloading to C:\Users\user\.cache\kagglehub\datasets\lakshmi25npathi\imdb-dataset-
of-50k-movie-reviews\1.archive...

100%|██████████| 25.7M/25.7M [00:02<00:00, 12.1MB/s]
Extracting files...

Path to dataset files: C:\Users\user\.cache\kagglehub\datasets\lakshmi25npathi\imdb-
dataset-of-50k-movie-reviews\versions\1
```

In [7]:

```python
%pip install pandas
```

```
Collecting pandas
  Downloading pandas-3.0.5-cp313-cp313-win_amd64.whl.metadata (19 kB)
Requirement already satisfied: numpy>=1.26.0 in c:\Users\user\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from pandas) (2.4.4)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\user\AppData\Roami
ng\Python\Python313\site-packages (from pandas) (2.9.0.post0)
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Successfully installed pandas-3.0.5 tzdata-2026.4
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [8]:

```python
import pandas as pd

# Load the CSV file in a dataframe
df = pd.read_csv("IMDB Dataset.csv")

# Show the first few rows
df.head()
```

Out[8]:

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. <br /><br />The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

In [9]:

```python
df.describe()
```

Out[9]:

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's show!!! It was a variety and not...	positive
freq	5	25000

In [10]:

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   review     50000 non-null  str
 1   sentiment  50000 non-null  str
dtypes: str(2)
memory usage: 781.4 KB
```

In [11]:

```python
# Check class distribution
print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

```
In [ ]:
```